

CAUSAL INFERENCE IN FINANCIAL EVENT STUDIES

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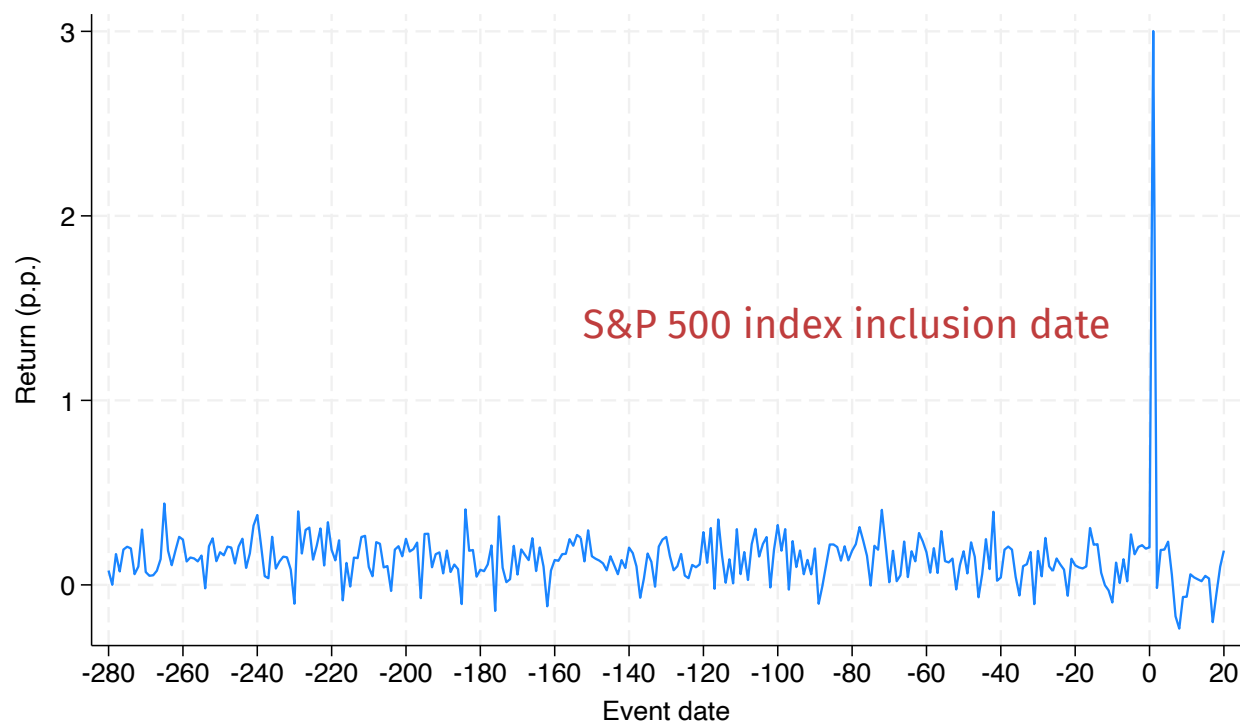
June 2026 <https://paulgp.com>

Package: <https://github.com/paulgp/financial-event-studies-package>

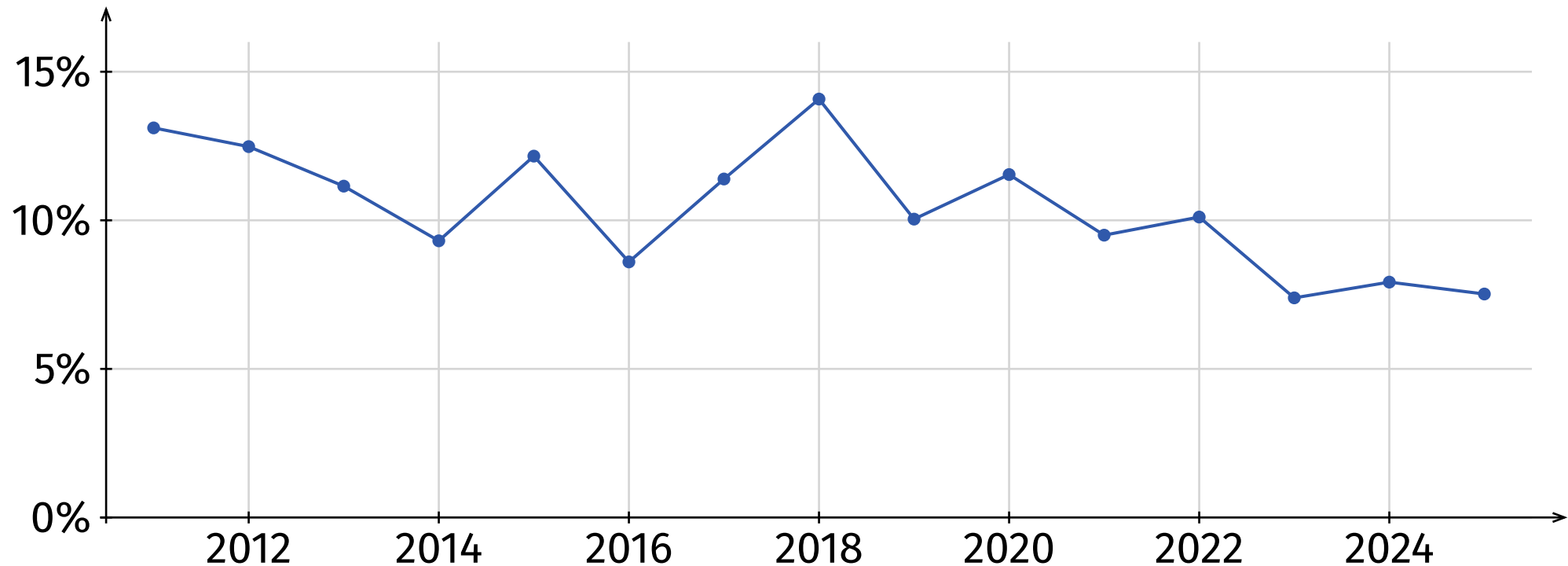
1. Finance literature studying the impact of **events** on **asset prices**
2. Econometrics literature estimating causal effects

What types of financial events? Examples...

- Index Inclusion
- Mergers and acquisitions
- IPO, SEO, Shares repurchased
- CEO/CFO Changes
- Patent Issuance
- Political events



% of JF / JFE / RFS articles mentioning financial event studies



Source: <https://paulgp.com/econlit-pipeline/search.html> — search of JF, JFE, RFS articles mentioning “cumulative abnormal return[s]” or “announcement return[s]” or “buy-and-hold return[s]”.

1. Financial event studies = **causal inference**
 - What is the counterfactual return?
2. When are abnormal return estimators biased?
 - Short-run, many events: **usually fine**
 - Short-run, one event: **depends**
 - Long-run: **almost always biased**
3. Propose alternative estimators
 - Synthetic control, PCA regression

Four applications:

1. Geithner / political connections
2. S&P 500 Index Inclusion
3. Mergers and acquisitions
4. **Close merger contests**

Some established findings reflect **model misspecification**, not treatment effects

- Unit of analysis: a path of stock returns $R_i = \{R_{i1}, \dots, R_{iT}\}$
 - Set of n securities (firms) observed over T time periods
 - T_i = when firm i 's event happens
 - $D_i = 1$ flags the $n_0 < n$ treated firms (single event: all $T_i = t_0$)
- For each stock and time period, there are two potential versions of R_{it} :
 - $R_{it}(1)$: the firm experienced the event
 - $R_{it}(0)$: the firm without the event
- We argue that researchers want the *causal effect* of the event:

$$\underbrace{\mathbb{E}(\tau_{it} \mid D_i = 1)}_{\text{ATT}} = \underbrace{\mathbb{E}(R_{it}(1) \mid D_i = 1)}_{\text{observed}} - \underbrace{\mathbb{E}(R_{it}(0) \mid D_i = 1)}_{\text{unobserved}}$$

- How can we approximate $\mathbb{E}(R_{it}(0) \mid D_i = 1)$?

Textbook approach with *abnormal returns* (Campbell, Lo, MacKinlay (1997))

$$AR_{it} = R_{it} - \underbrace{\mathbb{E}(R_{it} \mid X_t)}_{\text{Normal Returns given } X_t}$$

- $\mathbb{E}(R_{it} \mid X_t)$ can reflect many models of expected returns (MacKinlay (1997))
 - Market Model, CAPM, Fama-French

$$R_{i,t} = \alpha_i + \overbrace{\beta_i^o}^{\text{factor loading}} \underbrace{F_t^o}_{\text{priced (observed) factor}} + \underbrace{\tau_t D_i}_{\text{assume no effect before } t_0} + \varepsilon_{i,t}$$

1. Estimate α_i, β_i^o on pre-event data
2. Predict using the *realized* F_t^o
3. Average across events

$$\hat{\mathbb{E}}(R_{it}(0) \mid D_i = 1) \approx \hat{\alpha} + \hat{\beta} \hat{\mathbb{E}}(F_t)$$

Split returns into **priced** factors and **unpriced** factors:

$$R_{it} = \alpha_i + \beta_i^o F_t^o + \beta_i^u F_t^u + \varepsilon_{it}, \quad \mathbb{E}(F_t^u) = 0$$

AP model targets expected returns:

$$\mathbb{E}(R_{it}) = \alpha_i + \beta_i^o \mathbb{E}(F_t^o)$$

- F_t^u drops out — it earns no premium
- If model is estimated with many random events $\rightarrow F_t^o$ averages out too, so the model “works”

Counterfactual needs the contemporaneous return:

$$R_{it}(0) = \alpha_i + \beta_i^o F_t^o + \beta_i^u F_t^u$$

- Must match *every* systematic shock *that event day*
- Includes the unpriced $\beta_i^u F_t^u$ the model correctly omits

A model can be *right* for expected returns yet *wrong* for the counterfactual.

Subtract the fitted model from the realized return. For a treated firm:

$$\underbrace{\widehat{AR}_{it} - \tau_{it}}_{\text{bias}} \approx \underbrace{(\beta_i^o - \hat{\beta}_i^o) F_t^o}_{[(a) \text{ loading error}]} + \underbrace{\beta_i^u (F_t^u - \overline{F}_{\text{pre}}^u)}_{[(b) \text{ omitted factor}]}$$

- **(a) Loading error.** Uses the realized $F_t^o \rightarrow$ vanishes as $\hat{\beta}_i^o \rightarrow \beta_i^o$.
 - Timing correlated with a factor you modeled correctly is harmless.
- **(b) Omitted factor.** You net out only its mean $\overline{F}_{\text{pre}}^u$ (via the intercept $\hat{\alpha}_i$)
 - Does the event average of F_t^u equals $\overline{F}_{\text{pre}}^u$ in the limit?

Approximate the unobserved $\mathbb{E}(R_{it}(0) \mid D_i = 1)$ with a portfolio of control firms:

$$\hat{R}_t(0) = \sum_{j:D_j=0} w_j R_{jt}, \quad w \text{ chosen to match the treated path pre-event}$$

1. **Match the past** — pick weights so the portfolio tracks treated firms *before* the event
 2. **Project forward** — carry the same weights into the event window
 3. **Difference** — treated minus matched portfolio = the estimated **ATT**
- Matches the treated firms' **realized** path → captures **all** factor exposures
 - Key assumption: model stability across the event (approximation holds)
 - **Synthetic control** (convex weights) and **Gsynth** (PCA-learned factors) both do this

Abnormal return bias caused by $\beta_i^u (F_t^u - \bar{F}_{\text{pre}}^u)$ depends on event timing T_i :

- **Single event**
 - a single draw $F_{T_i}^u$ can cause issues unless perfectly modeled
- **Many events, non-random timing**
 - the event-window mean $\bar{F}_{\text{ev}}^u$ estimates $\mathbb{E}(F^u)$ **badly** $\rightarrow$ residual $\bar{\beta}^u (\bar{F}_{\text{ev}}^u - \bar{F}_{\text{pre}}^u)$
 - Long horizon issue: even 0.02%/day compounds to $\approx 5\%$ /yr
- **Many events, random timing or random assignment**
 - $\bar{F}_{\text{ev}}^u \rightarrow \mathbb{E}(F^u) = 0$ (**timing**) or $\bar{\beta}$ matches the population (**assignment**) $\rightarrow$ residual vanishes

- Abnormal returns specify F_t^o , estimate $\hat{\beta}_i^o \rightarrow \text{bias} = \underbrace{(\beta_i^o - \hat{\beta}_i^o) F_t^o}_{\text{loading gap}} + \underbrace{\beta_i^u F_t^u}_{\text{omitted unpriced}}$
- Synthetic control / gsynth match the treated firms' realized pre-event path
 - Implicitly match *all* loadings — priced and unpriced — not just the ones you name
 - The counterfactual carries the same $\beta_i^u F_t^u$ that day $\rightarrow$ both pieces cancel
- Consistent as $n_0, T \rightarrow \infty$ even without random timing or random assignment
 - Key assumption: counterfactual model is *stable* over event

- Acemoglu et al. (2016): Geithner's Treasury nomination leaks Nov 21, 2008 → connected firms jump
 - Pooled ATT across abnormal returns, synthetic control, gsynth, synthdid
- Paper compares connected vs. not *within* banks; we widen the control group
- Key features:
 - **Single event**
 - **Unusual timing (financial crisis)**
 - **Short horizon**



Contents lists available at [ScienceDirect](#)

Journal of Financial Economics

journal homepage: www.elsevier.com/locate/jfec



The value of connections in turbulent times: Evidence from the United States[☆]

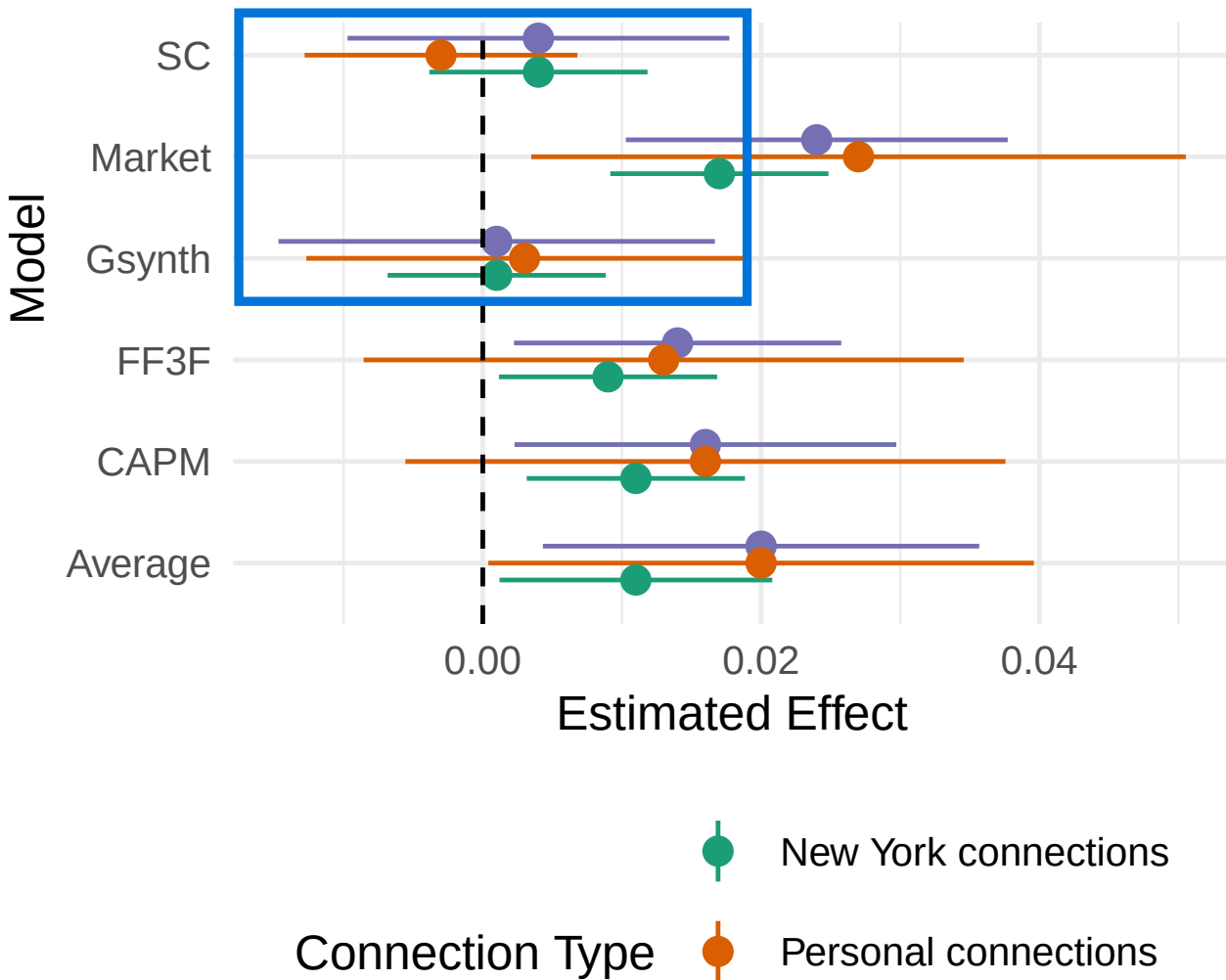
Daron Acemoglu^a, Simon Johnson^b, Amir Kermani^c, James Kwak^d, Todd Mitton^{e,*}

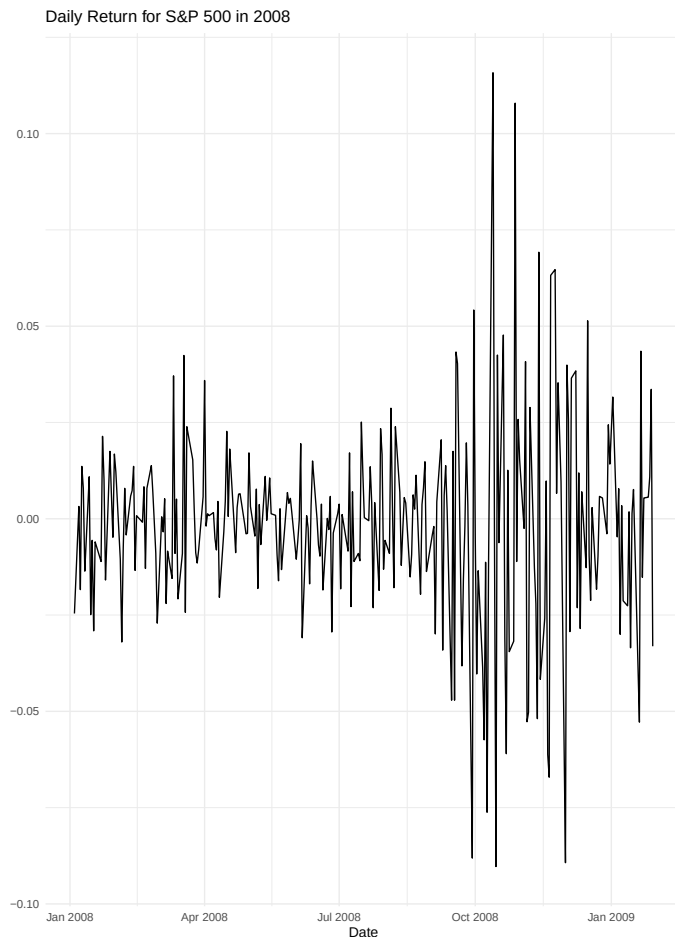
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ARTICLE INFO	ABSTRACT
Article history: Received 28 May 2015 Revised 14 September 2015 Accepted 12 October 2015 Available online 27 April 2016 JEL Classification: G01 G14 G21 G28 Keywords: Political connections Economic crises Institutions	The announcement of Timothy Geithner as nominee for Treasury Secretary in November 2008 produced a cumulative abnormal return for financial firms with which he had a prior connection. This return was about 6% after the first full day of trading and about 12% after ten trading days. There were subsequently abnormal negative returns for connected firms when news broke that Geithner's confirmation might be derailed by tax issues. Personal connections to top executive branch officials can matter greatly even in a country with strong overall institutions, at least during a time of acute financial crisis and heightened policy discretion.

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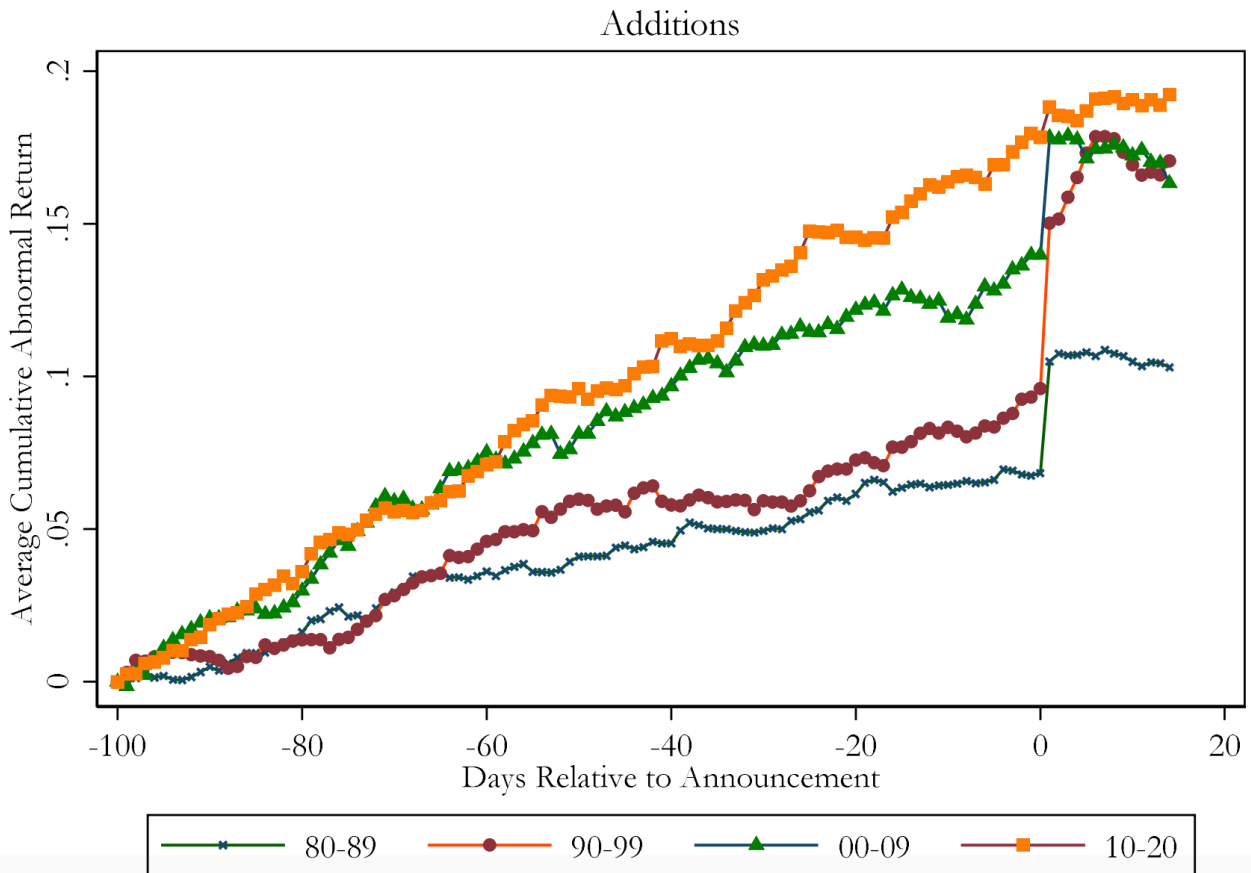
- Large effects in the paper shrink with synthetic control, even with banks-only controls
- Full control universe, all effects $\approx$ zero
 - Why?





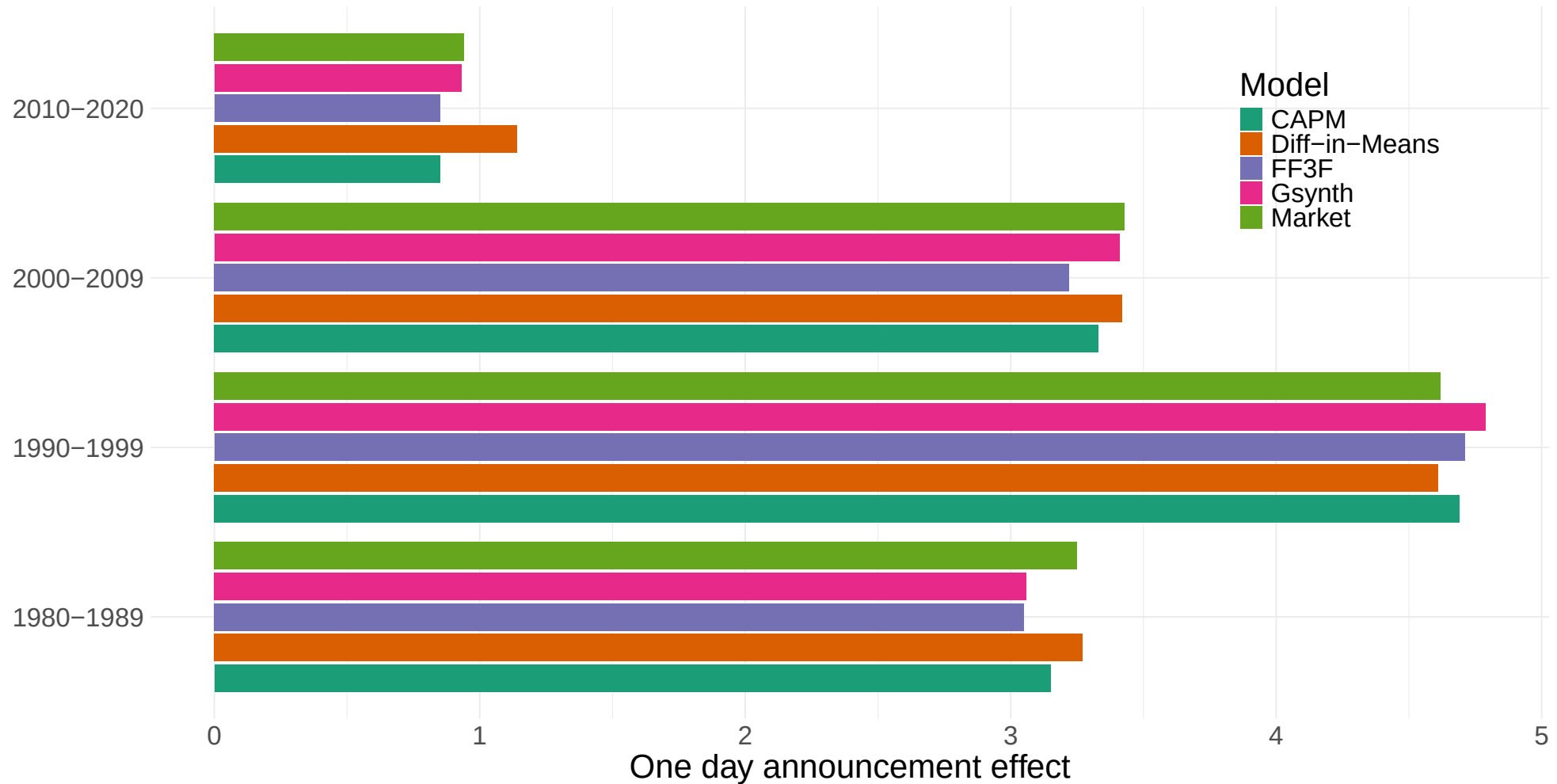
- Timing of event is correlated with significant risk factors
- S&P surged on Nov 21 — a market move in the extreme tail of its distribution
- Beta gap of $0.6 \times 6.9\%$ market move $\approx 4.1\%$ mechanical bias (one day)
- Synthetic methods match treated firms' loadings $\rightarrow$ effect disappears

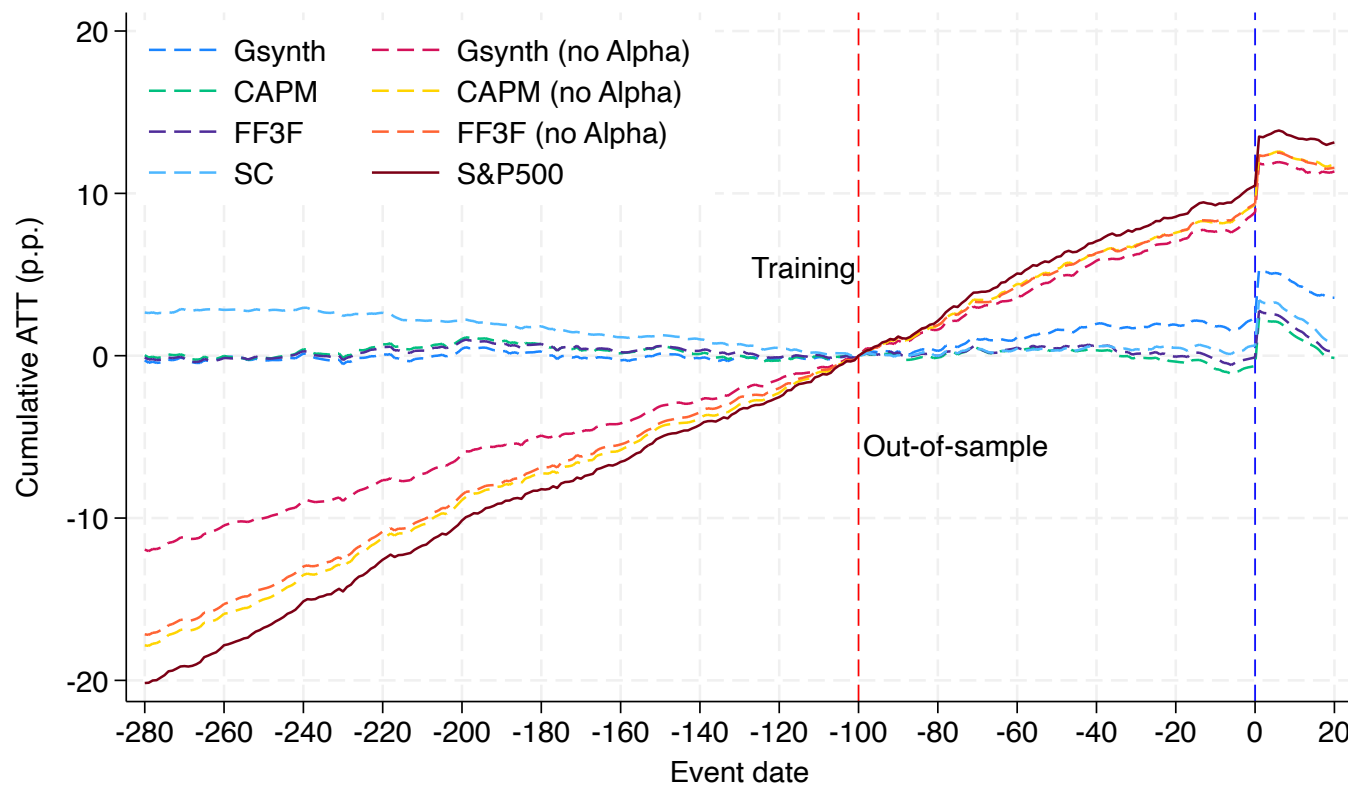
- Index inclusion effect: added firms jump on inclusion day
- Replicate Greenwood and Sammon (2025), S&P inclusions 1976-2020
- Key features:
 - Many events
 - As-if random timing
 - Short- and long-horizon



Short-run: method does not matter (folk wisdom holds)

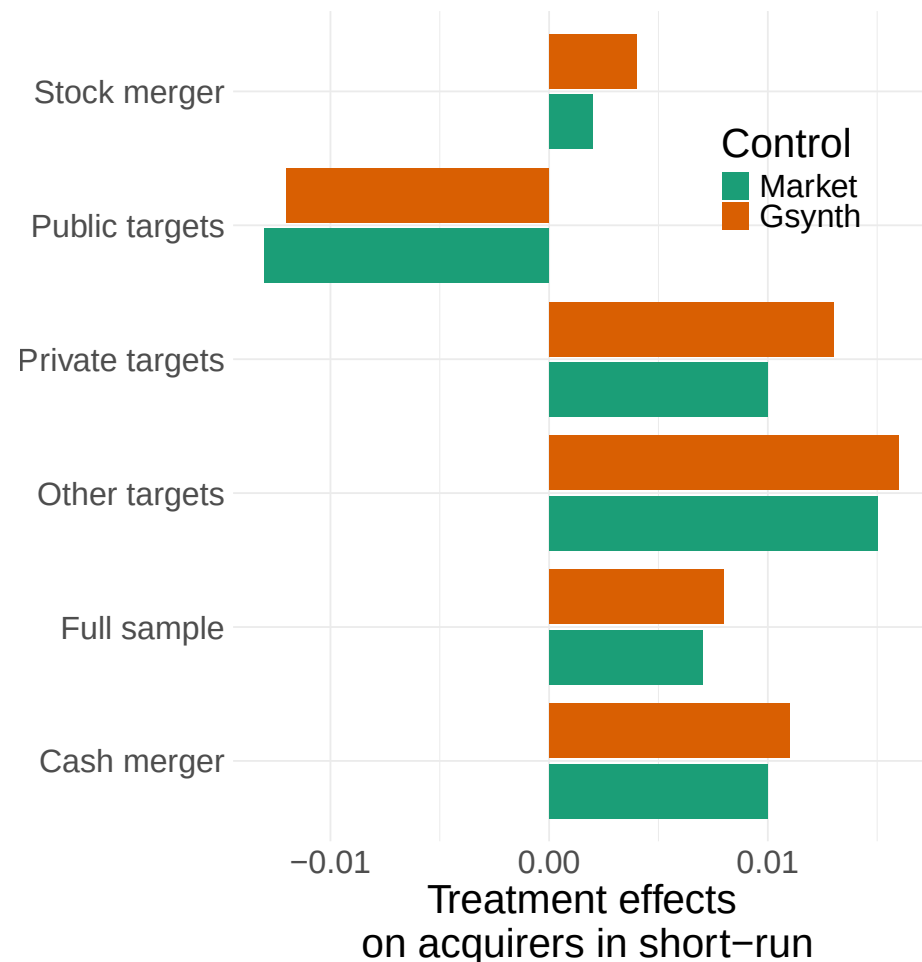
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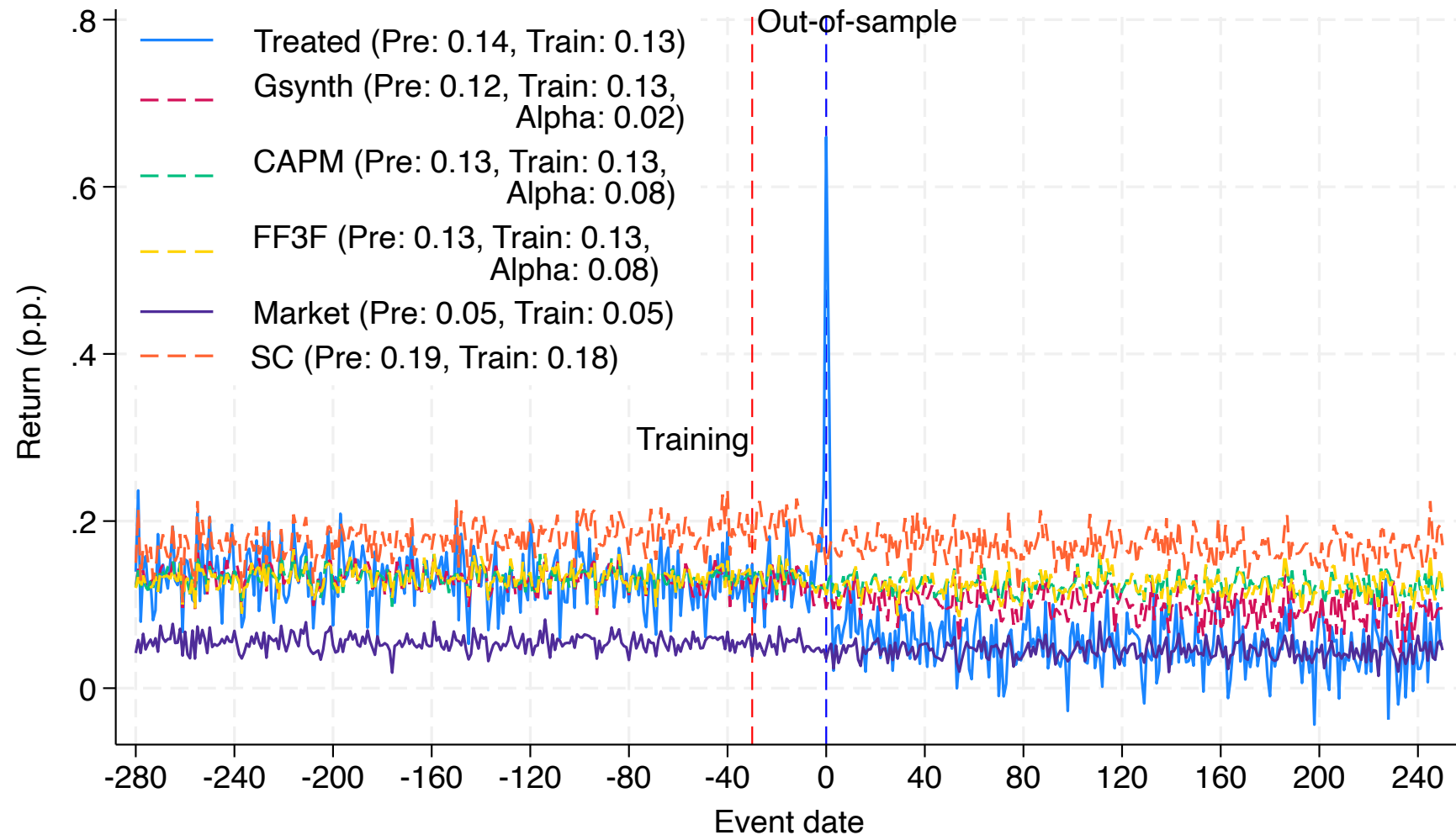


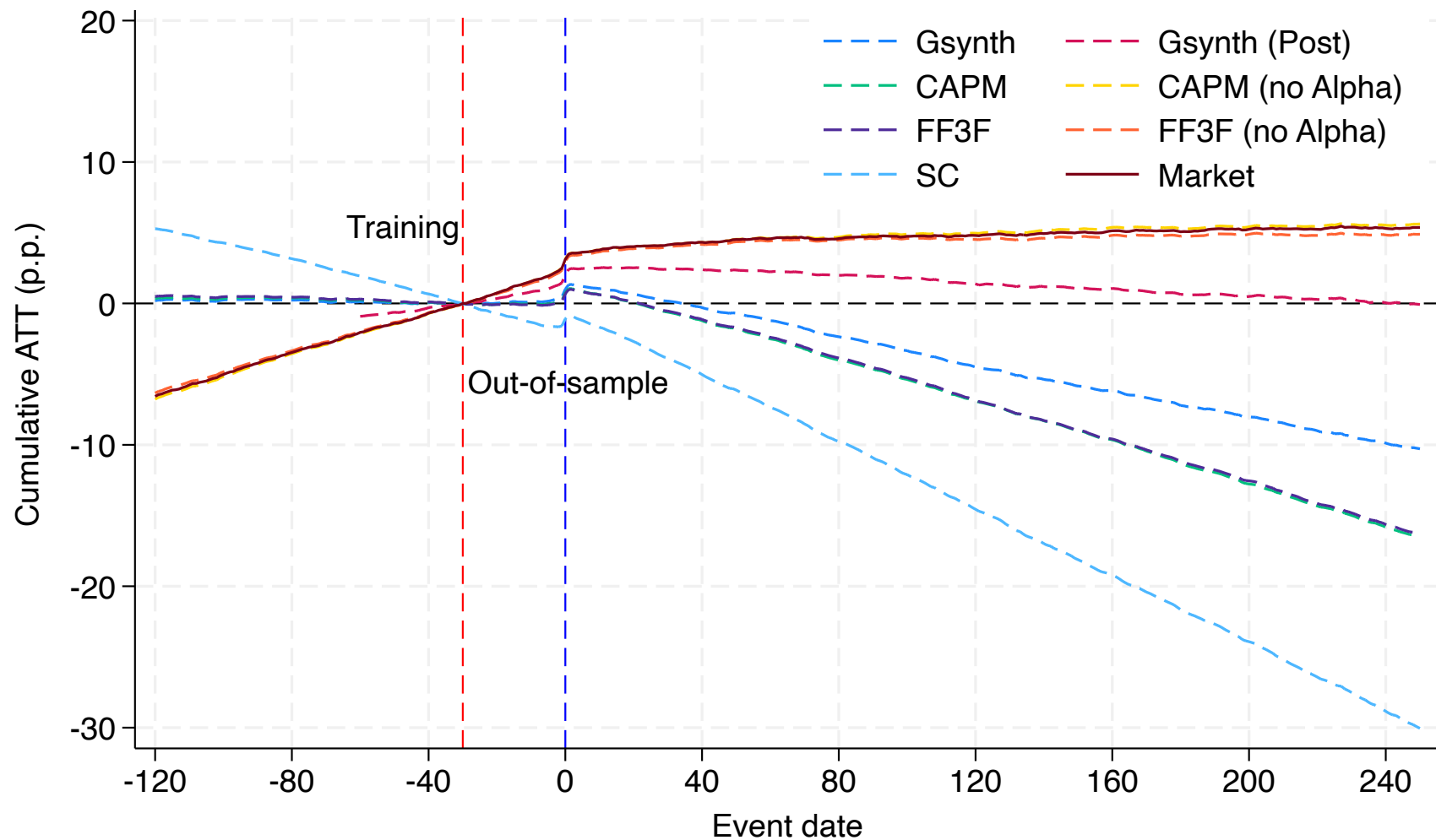


Without $\hat{\alpha}$, CAPM / FF3F / gsynth leave the drift; synthetic control (no intercept) matches the pre-period on its own.

- What is impact of acquisitions on *acquirer* returns?
- Replicate analysis from Malmendier (2017)
 - All acquisitions in SDC from 1980-2024
- Key features:
 - Many events
 - Unclear if timing is random (merger waves)
 - Short- and long- horizon
- Short-run effects quite similar, but...







The results were not materially different when returns were not corrected for market movements. — Shleifer (1986)

- Folk wisdom is **right** for short-run, many-event, random-timing studies
- **Wrong** for:
 - Single events during volatile periods (Geithner)
 - Long-horizon analyses (M&A, pre-inclusion drift)
 - Settings where treated firms have distinctive factor loadings
- Using expected return models for *contemporaneous* returns is the wrong tool for counterfactual prediction
 - Synth and Gsynth have benefit of known inference properties

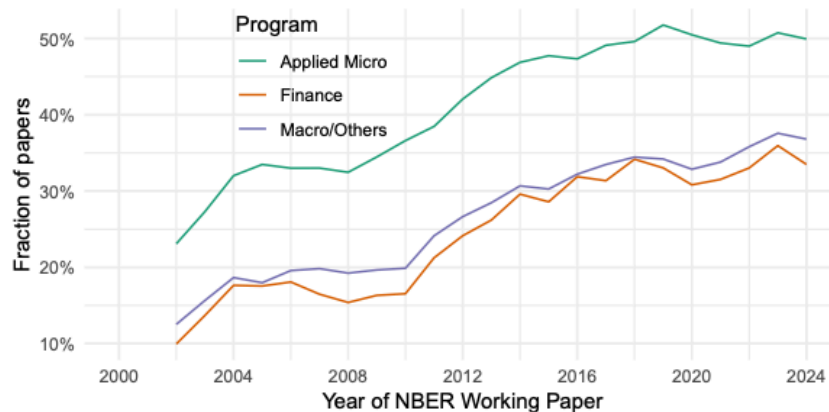
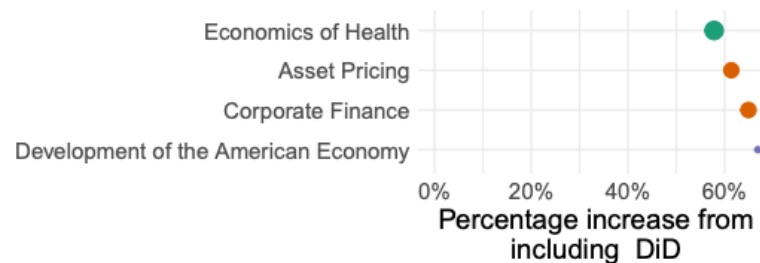
- Finance has used causal inference methods less
 - Yet owns data, experiments and outcomes other fields lack!
 - asset prices incorporate information faster than other outcomes

Take away:

- synthetic control / gsynth for single events and long horizons
- abnormal returns only for short-run, many-event, random-timing studies
- When you see an event study, ask: **“what is the counterfactual return for these firms?”**

Package & replication:

github.com/paulgp/financial-event-studies-package



(a) Identification

Goldsmith-Pinkham (2026)